How Tokens Are Calculated for Text, *Images* & Documents

AI ENGINEERING REFERENCE
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Prepared for course participants

Tokens are the fundamental unit of cost and context consumption in LLM APIs. The same content — text, an image, or a PDF — tokenizes very differently depending on modality and provider. Understanding these mechanics is essential for cost modelling, RAG system design, and context window management.

MODALITY 01

Text Tokens

Text is split into subword units by the model's tokenizer. A rough heuristic: **~4 characters ≈ 1 token** in English, but this varies by language, vocabulary, and model version.

RULE OF THUMB

1 token ≈ 4 chars ≈ ¾ of a word 750 words ≈ 1,000 tokens

Each provider uses a **different tokenizer**. The same text produces different counts across Claude, GPT, and Gemini. Opus 4.7 introduced a new tokenizer producing 1.0–1.35× more tokens than Claude 4.6 for identical input.

~4	1.35×
CHARS / TOKEN	OPUS 4.7 INFLATION

MODALITY 02

Image Tokens

Images are resized, then converted to tokens based on pixel area. The two major providers use fundamentally different formulas.

CLAUDE

tokens ≈ (w × h) / 750 Max long edge: 1568px (Opus 4.7: 2576px)

OPENAI HIGH DETAIL

tokens = 85 + (tiles × 170) Tiles are 512×512px after resize to 768px short edge

OpenAI also offers a **"low" detail mode** — flat 85 tokens regardless of size. GPT-4.1 mini/nano use 32×32 patches capped at 1,536 tokens.

~314	~1568
SMALL IMAGE	MAX (NON-OPUS 4.7)

MODALITY 03

PDF & Document Tokens

This is the expensive one. When Claude processes a PDF, it **extracts the text AND renders every page as an image**. You pay for both layers simultaneously.

CLAUDE PER PAGE

1,500–3,000 text tokens + page-as-image tokens 50 pages → 75K–150K tokens total

Documents *with* embedded images (charts, photos) have no separate line item — the visuals are captured in the page-image rendering. Pre-extracting text and sending it as plain text can **cut token cost by 50%+**.

~2K	60,934
PER PAGE (AVG)	30-PG PDF (OPUS 4.7)

Provider Comparison — Image Token Examples

IMAGE SIZE	CLAUDE (SONNET 4.6)	CLAUDE (OPUS 4.7)	OPENAI GPT-4O (HIGH)	OPENAI GPT-4.1 MINI
682 × 318	~310 tokens	~314 tokens	85 + 170×2 = 425	~289 patches
1024 × 1024	~1,398 tokens	~1,398 tokens	85 + 170×4 = 765	~576 patches
2048 × 2048	~1,568 (capped)	~4,784 tokens	85 + 170×12 = 2,125	~1,536 (capped)
3456 × 2234	~1,568 (capped)	~4,784 tokens	85 + 170×12 = 2,125	~1,536 (capped)

⚡ Practical Implications for RAG

- **Chunk sizing:** ~500 text tokens ≈ 1 paragraph, but a single image costs 2–3 chunks worth of tokens.
- **PDF pre-processing:** Extract text with `pymupdf` / `pdfplumber` before sending to an LLM. Only send page images if visual layout matters.
- **Cost modelling:** 5 text chunks (~2,500 tokens) vs. 5 chunks + 3 page images (~7,000+ tokens). Budget accordingly.
- **Pre-flight check:** Anthropic's `count_tokens` endpoint is free — use it before committing to expensive calls.
- **Caching:** Use prompt caching (`cache_control: ephemeral`) for PDFs queried repeatedly — pays for itself after 2–3 calls.

📚 Reference Links

- → **Claude Vision Docs** — image formulas, resize rules, limits
  platform.claude.com/docs/…/vision
- → **Claude PDF Support** — dual-channel processing, cost per page
  platform.claude.com/docs/…/pdf-support
- → **Claude Token Counting API** — free pre-flight endpoint
  platform.claude.com/docs/…/token-counting
- → **OpenAI Images & Vision** — tile formulas, detail modes
  platform.openai.com/docs/guides/images-vision
- → **Simon Willison's Token Counter** — model comparison tool
  simonwillison.net/…/claude-token-counts
- → **Claude Tokenizer Tool** — interactive token counter
  claudetokenizer.com

Key insight: **the same content costs different tokens on different models** — always measure, don't assume.